SCORE/30:

Math 171 (sec. 501)
Analytic Geometry & Calculus
Monday, January 31, 2008

NAME: ____________________

PRACTICE QUIZ 3 (with solutions)

Please show your work and write only in **pen**. Calculuators are **not** allowed. Notes are **not** allowed. Please also note that there are exactly **2** problem.

1: A parrot  sitting at the point $(4,5)$ on the line L defined by $9x-7y=1$ decides to start flying away at time $t=11$, always along a line perpendicular to L.
Please find an explicit vector function $\vec{r}:[0,\infty)\longrightarrow\mathbb{R}^2$ that traces out the parrot's path.

The basic idea is to first find a vector $\vec{n}$ perpendicular to L and then add a suitable multiple of $\vec{n}$ to $(4,5)$.

In particular, from class, we know that $\vec{n}=(9,-7)$ is perpendicular to L. We can then simply form

$$\boxed{\vec{r}(t):=(4,5)+(t-11)(9,-7)}$$

(or $\vec{r}(t)=(4+(t-11)9,5+(t-11)(-7))=(9t-95,-7t+82)$), which indeed satisfies $\vec{r}(11)=(9,-7)$ and has image forming a line perpendicular to L. (To see the latter, one can simply observe that the displacement vector $\vec{r}(t)-\vec{r}(11)$ is always parallel to the normal vector $\vec{n}$.)

2: Please fill in the blanks:
$\lim\limits_{x\to a} f(x)=L \iff$ for every $\varepsilon>0$ there is a $\delta>0$ such that ____________________.

$$\boxed{\lim_{x\to a} f(x)=L \iff \text{for every } \varepsilon>0 \text{ there is a } \delta>0 \text{ such that } |f(x)-L|<\varepsilon \text{ whenever } 0<|x-a|<\delta.}$$

3:

(a) Please write explicitly, using ε and δ as in the definition of limit, the condition that $\lim\limits_{x\to 0} \sin(1/x)$ is **not** defined.
(b) Using your answer from (a), prove rigorously that $\lim\limits_{x\to 0} \sin(1/x)$ is **not** defined.

The key is to take the formal definition of limit, apply it to the case at hand, and **negate** it. First, one should know some background:

1. We let $\neg$ denote logical negation ("not").
2. Negating a quantified sentence flips the quantifier and negates the sentence. For example, the negation of "All Aggies are honest." is "Some Aggie is dishonest." In symbols,

$$\neg\forall x \in \{\text{Aggies}\}\ [x \text{ is honest.}] \;=\; \exists x \in \{\text{Aggies}\}\ [x \text{ is } \textbf{dis}\text{honest.}]$$

3. Put another way, there are two simple rules to remember:

$$\text{(i) } \neg\forall x\ P(x) = \exists x\ \neg P(x), \quad \text{(ii) } \neg\exists x\ P(x) = \forall x\ \neg P(x)$$

Now, back to the problem: The answer to Part (a), in jargon, is

$$\neg\left(\exists L\in\mathbb{R}\left[\lim_{x\to 0}\sin(1/x) = L\right]\right).$$

So to get this into English, we insert the definition of limit and then do our negation tricks:

$$\neg\,(\exists L\in\mathbb{R}\ \forall\varepsilon>0\ \exists\,\delta>0\ \forall x\in\mathbb{R}\ \ [|\sin(1/x)-L|<\varepsilon \text{ whenever } 0<|x|<\delta.])$$

$$\forall L\in\mathbb{R}\ \neg\,(\forall\varepsilon>0\ \exists\,\delta>0\ \forall x\in\mathbb{R}\ \ [|\sin(1/x)-L|<\varepsilon \text{ whenever } 0<|x|<\delta.])$$

$$\forall L\in\mathbb{R}\ \exists\varepsilon>0\ \neg\,(\exists\,\delta>0\ \forall x\in\mathbb{R}\ \ [|\sin(1/x)-L|<\varepsilon \text{ whenever } 0<|x|<\delta.])$$

$$\forall L\in\mathbb{R}\ \exists\varepsilon>0\ \forall\,\delta>0\ \neg\,(\forall x\in\mathbb{R}\ \ [|\sin(1/x)-L|<\varepsilon \text{ whenever } 0<|x|<\delta.])$$

$$\forall L\in\mathbb{R}\ \exists\varepsilon>0\ \forall\,\delta>0\ \exists x\in\mathbb{R}\ \neg\,[|\sin(1/x)-L|<\varepsilon \text{ whenever } 0<|x|<\delta.]$$

$$\forall L\in\mathbb{R}\ \exists\varepsilon>0\ \forall\,\delta>0\ \exists x\in\mathbb{R}\ \ [|\sin(1/x)-L|\geq\varepsilon \text{ and } 0<|x|<\delta.]$$

So, in English, we get:

$\boxed{\text{For all real } L\text{, there is an } \varepsilon>0 \text{ such that for any } \delta>0\text{, we can find a real } x \text{ with } |\sin(1/x)-L|\geq\varepsilon \text{ and } 0<|x|<\delta.}$

For Part (b), we'll need to actually **prove** the last boxed sentence. Here, we break into 4 exclusive cases:

$\boldsymbol{L>1}$**:** Here, we can simply take $\varepsilon = \frac{L-1}{2}$. For then, $|\sin(1/x) - L| > \varepsilon$ for **all** x, since the sin function has range in $[-1,1]$. So we certainly have $|\sin(1/x) - L| > \varepsilon$ when $0 < |x| < \delta$ for any $\delta > 0$.

$\boldsymbol{L<-1}$**:** Almost identical to the last case, except we take $\varepsilon = \frac{-L-1}{2}$.

$\boldsymbol{L \in [-1,1] \setminus \{0\}}$**:** Take $\varepsilon = L/2$ and observe that for any $\delta > 0$, we can always find an x with $0 < x < \delta$ and $1/x$ a multiple of π. In particular, for such an x, we have $\sin(1/x) = 0$, and thus $|\sin(1/x) - L| = |L| > \varepsilon$.

$\boldsymbol{L=0}$**:** Here we can take $\varepsilon = 1/2$ and observe that for any $\delta > 0$, we can always find an x with $0 < x < \delta$ and $1/x$ equal to $\pi/2$ plus a multiple of π. In particular, for such an x, we have $\sin(1/x) = 1$, and thus $|\sin(1/x) - L| = 1 > \varepsilon$.

So we are done.

4:

(a) Please sketch the graph of a function f that satisfies the following properties:

(i) $\lim_{x \to -\infty} f(x) = 2$

(ii) $\lim_{x \to -1^-} f(x) = +\infty$

(iii) $\lim_{x \to -1^+} f(x) = -\infty$

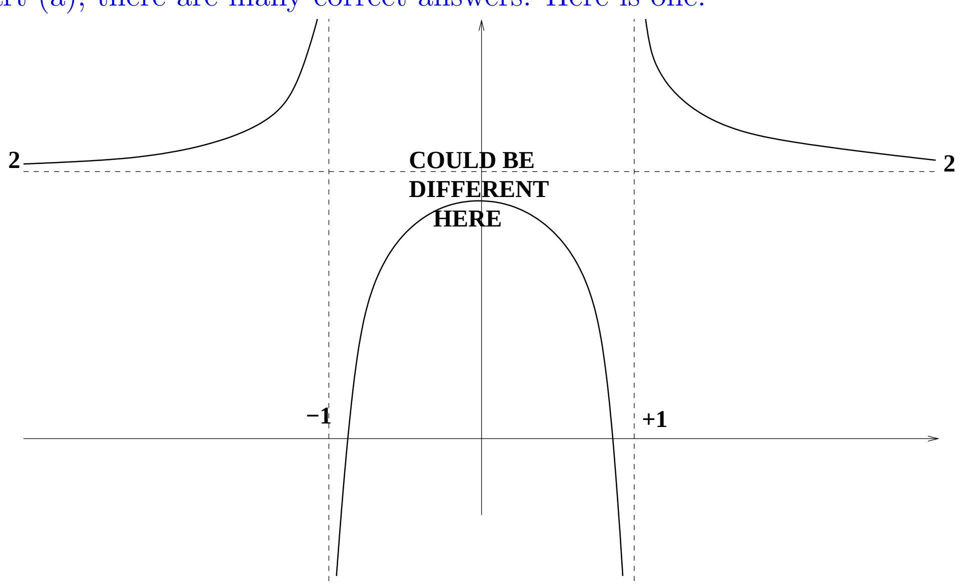
(iv) $\lim_{x \to 1^-} f(x) = -\infty$

(v) $\lim_{x \to 1^+} f(x) = +\infty$

(vi) $\lim_{x \to 1^+} f(x) = 2$

(b) Please find a rational function whose graph satisfies the conditions from Part (a).

For Part (a), there are many correct answers. Here is one:



For Part (b), we need to actually find a rational function with a graph resembling the picture above. To do this, we can first try to find rational functions that blow up in the right way at -1 and 1.

From class, we know we can easily modify the function $x \mapsto 1/x$ to get what we need. In particular, note that $g(x) := -1/(x+1)$ satisfies Conditions (i)–(iii), and that $h(x) := 1/(x-1)$ satisfies Conditions (iv)–(vi). Since, for any small $\varepsilon > 0$, g is bounded on $[-1+\varepsilon, 1]$ and h is bounded on $[-1, 1-\varepsilon]$, we can then try to add the two functions.

This turns out to work: Defining

$$f(x) := g(x) + h(x) = \frac{-1}{x+1} + \frac{1}{x-1} = \frac{x+1-(x-1)}{x^2-1} = \frac{2}{x^2-1},$$

we see that all of Conditions (i)–(vi) are satisfied. (Exercise: Check all 6 conditions for this f!) So $\boxed{f(x) := \frac{2}{x^2-1}}$ works.

5:

Please find a number c such that $\lim_{x\to+\infty} \left(\sqrt{x^2+cx} - x\right) = 5$.

Here, we have the limit of a difference of quantities that tend to $+\infty$. For such problems, one always tries to combine the difference into something more managable. Here, the square root should remind you of the simplification by conjugates you've seen in high school, e.g., $\frac{1}{7-\sqrt{3}} = \frac{7+\sqrt{3}}{(7-\sqrt{3})(7+\sqrt{3})} = \frac{7-\sqrt{3}}{7-3} = \frac{7-\sqrt{3}}{4}$. So let's first try to compute the limit, as a function of c, and then solve for the right c.

It thus makes sense to proceed as follows:
$\lim_{x\to+\infty} \left(\sqrt{x^2+cx} - x\right) = \lim_{x\to+\infty} \frac{(\sqrt{x^2+cx}-x)(-\sqrt{x^2+cx}-x)}{-\sqrt{x^2+cx}-x} = \lim_{x\to+\infty} \frac{x^2-(\sqrt{x^2+cx})^2}{-\sqrt{x^2+cx}-x} = \lim_{x\to+\infty} \frac{-cx}{-\sqrt{x^2+cx}-x} = \lim_{x\to+\infty} \frac{cx}{\sqrt{x^2+cx}+x} = \lim_{x\to+\infty} \frac{c}{\sqrt{1+c/x}+1} = c/2.$

So setting $\boxed{c = 10}$ works.